GenerChange: Uncertainty-Aware Domain Generalization Network for Cross-Domain Remote Sensing Change Detection

Leyuan Fang, *Senior Member, IEEE*, Runzhe Wang, Xin Yi, Shaobo Xia, Shuhe Wang, and Yangbo Feng

***Abstract*—Remote sensing change detection (RSCD) aims to identify changes in ground objects at the same geographic location across dual-temporal images. Despite deep learning's success in this field, most existing methods fail to generalize from a source domain (seen) to a target domain (unseen) without retraining. Cross-domain RSCD can learn a transferable change detection model in the source domain, enabling target-domain change detection without retraining. However, existing methods ignore the stylistic variations between dual-temporal images and the lack of prior knowledge about the target domain, which together lead to increased model aleatoric and epistemic uncertainty. To address the above issues, we propose an uncertainty-aware domain generalization network (GenerChange) for cross-domain RSCD. It employs a dual-covariance style-debias mechanism and an evidence-aware uncertainty estimation module to reduce model uncertainty. First, the dual-covariance style-debias mechanism can effectively disentangle content features from style variations, reducing aleatoric uncertainty by aligning the covariance between the enhanced and original images. Furthermore, the evidence-aware uncertainty estimation module models the Dirichlet distribution of pixel change probabilities. This helps the model focus on domain-invariant information during training, which can be generalized to the target domain, enabling the model to reduce the epistemic uncertainty. Extensive experimental results demonstrate that the proposed method outperforms existing domain generalization (DG) methods for RSCD.**

***Index Terms*—Domain generalization (DG), remote sensing change detection (RSCD), uncertainty estimation.**

I. Introduction

REMOTE sensing change detection (RSCD) aims to identify and locate ground change information by analyzing multitemporal remote sensing images [1]. This enables the effective identification of land cover transformations. It has been applied in a wide range of fields, including natural disaster assessment [2], [3], [4], and environmental monitoring [5], among others. Initial methodologies [6], [7] typically employed pixel-wise differencing to construct change maps, followed by threshold-based segmentation to isolate modified regions [8]. Such approaches primarily rely on low-level feature hierarchies with hand-crafted architectures, constraining discriminative feature extraction from raw data. This structural deficiency leads to inadequate representation learning capacity, ultimately hindering robustness in practical applications. Although these techniques succeeded in detecting change objects, they struggled to differentiate meaningful semantic shifts from incidental variations. Recently, many convolutional neural networks (CNNs) [9], [10], [11] and transformers [12], [13], [14] have shown great results in RSCD. However, traditional RSCD models mentioned above are typically trained and tested within the same source domain, making it difficult to generalize to different target domains due to domain gaps. In remote sensing data, domain gaps can be caused by different data distributions of different RSCD datasets, such as differences in scenes, illumination conditions, and image conditions. For example, a model trained on densely packed villas in affluent residential areas may fail to generalize when encountering unseen factory structures in suburban regions. These structures often have sparser distributions and distinct roof textures. Therefore, in the RSCD task, the change features in the source domain can significantly differ from the target domains, presenting a huge challenge in generalization.

To address this challenge, domain adaptation RSCD methods [15], [16], [17], [18] are proposed. These methods employ domain alignment strategies to align the distributions of the source and target domains, which helps the model to bridge the gap between the source and target domains. The domain adaptation RSCD methods are typically trained on the source domain and fine-tuned with target-domain information to adjust the model's distribution. Specifically, DSDANet [15] was first developed to improve cross-domain performance in RSCD, which embeds feature representation into a reproducing kernel Hilbert space to learn domain-invariant features. Qu et al. [17] propose a homogeneous mutual representation network that combines a graph feature alignment module to improve generalization ability. In addition, to further improve the generalization ability of the model, Xie et al. [18] propose three co-learning combination modules that use digital geographic information as additional inputs to transfer mutual information between modalities. However, a significant

Received 12 May 2025; revised 27 August 2025; accepted 26 September 2025. Date of publication 1 October 2025; date of current version 30 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62425109, Grant U22B2014, and Grant 42201481; and in part by the Science and Technology Plan Project Fund of Hunan Province under Grant 2022RC3064. *(Corresponding author: Yangbo Feng.)*

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Digital Object Identifier 10.1109/TGRS.2025.3616443

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limitation of domain adaptation is that it needs to use target-domain information for auxiliary training during the training phase. In practical applications, it is difficult to obtain data from the target domain to train a model.

To overcome the limitations of domain adaptation, domain generalization (DG) [19], [20], [21], [22], [23], [24] has been introduced. Compared with domain adaptation, DG does not require target-domain information for auxiliary training. Specifically, previous work [25] applies instance normalization layers to standardize features, thereby improving the model's generalization ability. Zhou et al. [19] develop an efficient data augmentation approach to expand the types of the source domain by generating samples in the training phase. Lee et al. [26] propose a semantic alignment strategy to learn domain-generalized semantic information from unrelated data. Furthermore, a cyclic-disentangled self-distillation method [24] is proposed to cyclically extract domain-invariant representations from single-source domain data. Although these methods achieve impressive performance, they are limited in their application domains. They are mainly applied to semantic segmentation [19], [20], [21], [22], [23] and object detection [24], while research on the RSCD task remains limited.

There are two issues that hinder the direct application of the above DG methods to RSCD as follows.

1) Most DG approaches are designed for single-image detection and ignore the style differences between dual-temporal images. The style differences between dual-temporal images introduce more aleatoric uncertainty, which impacts the model's generalization to target domains. The aleatoric uncertainty is caused by the uncertain data distribution, such as random style variations across different images. For example, when detecting building changes over time, some stylistic variations may significantly affect the results, such as lighting, climate, and seasonal variations. Although the building structures are unchanged, the roof textures of the building may be completely different between summer and winter images. These style differences in dual-temporal images increase the aleatoric uncertainty for distinguishing changes.

2) The lack of target-domain images increases epistemic uncertainty [27] during the training phase. The epistemic uncertainty is caused by the model's lack of sufficient different distributions. This results in the model being unable to distinguish between domain-specific and domain-invariant representations in DG RSCD. The domain-specific representations are typically related to the unique attributes of a domain, such as seasonal variations, lighting conditions, and shadow changes. In contrast, domain-invariant representations are universally applicable to different domains, such as high-level semantic information of buildings. Existing RSCD methods [9], [12], [13] focus on learning domain-specific representations and lack generalization ability for target domains. This is because the training data is limited to the source domain, and the model cannot learn domain-specific information for the target domains. When the model overly focuses on domain-specific features and neglects domain-invariant features, it leads to insufficient understanding of target domains, increasing the epistemic uncertainty of the model and impacting the cross-domain RSCD performance.

There are numerous methods for estimating the model uncertainty, Bayesian neural networks [28], [29], and the deep ensemble algorithm [30], [31] estimate the uncertainty by adding an extra module. This introduces a high cost of inference time and cannot optimize the model instantly. To overcome limitations, deterministic uncertainty methods [32], [33], which can estimate the uncertainty in a single forward pass, are proposed. Specifically, Bao et al. [34] use additional uncertainty estimation to distinguish known and unknown actions in an open set task. In addition, ELFNet [35] is proposed to fuse the multilevel predictions and fuse features, CNN-based and transformer-based, for stereo matching. Chen et al. [36] propose a method to consider the dual-level uncertainties of different root sources to alleviate the interference of background noise for weakly supervised temporal action localization. Although these methods achieve great performance, they do not consider uncertainties brought about by DG problems, which is difficult to be applied to DG RSCD tasks.

These limitations motivate us to design a novel uncertainty estimation model for the DG RSCD task. The proposed approach, named uncertainty-aware DG network (GenerChange), addresses the aforementioned issues. Our GenerChange consists of the dual-covariance style-debias mechanism and the evidence-aware uncertainty estimation module.

1) The dual-covariance style-debias mechanism is designed to reduce aleatoric uncertainty brought by style differences between dual-temporal images. Specifically, this module mitigates the interference of irrelevant style variations in the feature space by aligning the covariance between original and augmented same temporal images, which share identical content but different styles. However, this process inevitably removes content information. Therefore, a dual-temporal covariance loss function is applied to maintain the semantic consistency of different temporal images. This mechanism effectively eliminates style-induced interference on content to reduce aleatoric uncertainty arising from dual-temporal image style changes.

2) The evidence-aware uncertainty estimation module is applied to reduce epistemic uncertainty by leveraging evidence deep learning (EDL) [37], which addresses the issue arising from the absence of target-domain images. Concretely, it predicts the Dirichlet distribution of pixel change probabilities, which can be regarded as a context evidence collection process. Different from traditional Gaussian mixture distribution modeling, the Dirichlet distribution can capture the correlation between dual-temporal image knowledge of different domains. This helps the model focus on domain-invariant information during training, which can be generalized to target domains. Moreover, the learned evidence from the Dirichlet distribution is informative in quantifying the predictive uncertainty of diverse change detection image types. If the model has low epistemic uncertainty, the reliability of the model remains high even when detecting changes in dual-temporal images of target domains. This alleviates the problem of knowledge differences in different domains.

Our contributions can be summarized as follows.

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1) A novel GenerChange is proposed for DG RSCD. To the best of our knowledge, the method is the first study to apply uncertainty estimation to the DG of RSCD, which shows that uncertainty estimation benefits in this task.

2) By carefully designing the dual-covariance style-debias mechanism, the proposed method can adaptively disentangle content and style features in dual-temporal images, reducing the aleatoric uncertainty.

3) We propose the evidence-aware uncertainty estimation module to alleviate the problem of knowledge differences in different domains, avoiding high epistemic uncertainty.

4) The proposed framework performs favorably against state-of-the-art methods on four standard datasets, which verify its effectiveness and generalization ability.

The rest of the article is organized as follows. Section II reviews the related work. We present the detailed configuration of the proposed framework and the optimization process in Section III. The experimental results are reported and analyzed in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Work

This section discusses existing works closely related to our research, focusing on RSCD, uncertainty estimation, and DG. We emphasize the challenges of cross-scene RSCD and the integration of uncertainty-aware mechanisms to address domain shift.

A. Remote Sensing Change Detection

RSCD has been extensively studied and applied [6], [7], [14], [38], [39], [40], [41], [42]. RSCD aims to identify semantic changes between dual-temporal remote sensing images. Traditional early change detection methods [6], [7] often involved generating change maps through pixel-level difference analysis, followed by thresholding to determine the changed areas [8]. Although these methods can reveal change information to some extent, they are limited in distinguishing semantic changes from irrelevant changes due to the limitations of RGB-based features, especially when handling complex scenes and subtle changes. The effectiveness of these traditional methods is often limited. Therefore, despite their widespread use in the early days, with technological advancements, researchers have begun to seek more advanced techniques to improve the accuracy (Acc) and efficiency of change detection. In recent years, deep learning methods based on deep neural networks [9], [10], [11], [43] have shown significant performance improvements in many tasks, such as remote sensing image semantic segmentation [44], semantic segmentation labeling [45], hyperspectral image classification [46], [47], [48], object detection [49], and even video classification [50]. In the field of RSCD, various deep learning-based methods have also been explored [14], [38], [39], [40], [41], [42]. For example, Fang et al. [39] propose the Changer model, which deeply considers the interaction between dual-temporal features showing good results. Fang et al. [39] configure a weakly supervised method with the point label to detect changes and achieve highly competitive results.

With the advancement of transformer-based architectures, researchers have found that they can better capture global information, providing new opportunities to improve deep models. Consequently, transformers have been gradually incorporated into RSCD applications. Zhang et al. [40] introduce a pure transformer network for CD, which employs a twin U-shape structure of Swin transformer blocks to effectively extract multiscale features and capture global spatiotemporal information. However, the transformer architecture can have issues with preserving details, and thus Feng et al. [41] introduce a co-attention module to enhance features representation for better detection of change areas. Chen et al. [14] propose a BIT-CD model composed of CNN and transformer to capture the temporal and spatial context. In addition, Feng et al. [42] propose a novel intrascale and interscale feature fusion method using transformer modules. This method employs two weight-sharing backbone networks to extract different features from two images and then promotes information aggregation between single-temporal phase features and scale features of different backbones. As summarized by Liu et al. [51], the diffusion model is used in the RSCD task and achieves great results. However, the aforementioned methods only focus on the performance of a single source domain and often show significant performance degradation when faced with cross-domain scenarios in applications.

B. Uncertainty Estimation and Optimization

Uncertainty quantification is critical for reliable decision-making in safety-sensitive applications. As deep learning technology is increasingly applied to real-world security-sensitive scenarios, measuring confidence becomes crucial in helping to distinguish suspicious predictions and assisting in decision-making [52]. Bayesian neural networks [53] enable deep models to possess uncertainty by replacing deterministic weight parameters with distributions. However, the computational cost of optimizing such a large number of parameters is prohibitive. Monte Carlo dropout (MC dropout) [29] is the most famous method to address this issue, formulating dropout as a random variable of a Bernoulli distribution to approximate the training process as variational inference.

Deep ensemble methods [30], [31], [54] predict overall results based on multiple models with different architectures. Furthermore, deterministic neural networks [32], [33], [55] provide a computationally efficient alternative, estimating predictive uncertainty through a single forward pass. Examples include evidential deep learning [56] and prior networks [57]. Bao et al. [34] propose evidential deep learning for open and closed-set action recognition tasks, using uncertainty scores as a criterion for unknown classes in open-set identification. Chen et al. [36] focus on issues in weakly supervised temporal action localization, proposing an uncertainty-aware dual-evidential learning framework. Deep evidential regression [58] extends [56] to regression tasks, estimating the inverse gamma distribution parameters over a normal distribution, which ensures explicit representation of epistemic and aleatoric uncertainty. Considering a multimodal setup, Ma et al. [59] use a mixture of normal inverse gamma (MoNIG) distributions to allow a credible regression that characterizes the modality-specific and integrated uncertainty. Furthermore, Lou et al. [35] seek to integrate uncertainty estimation into the

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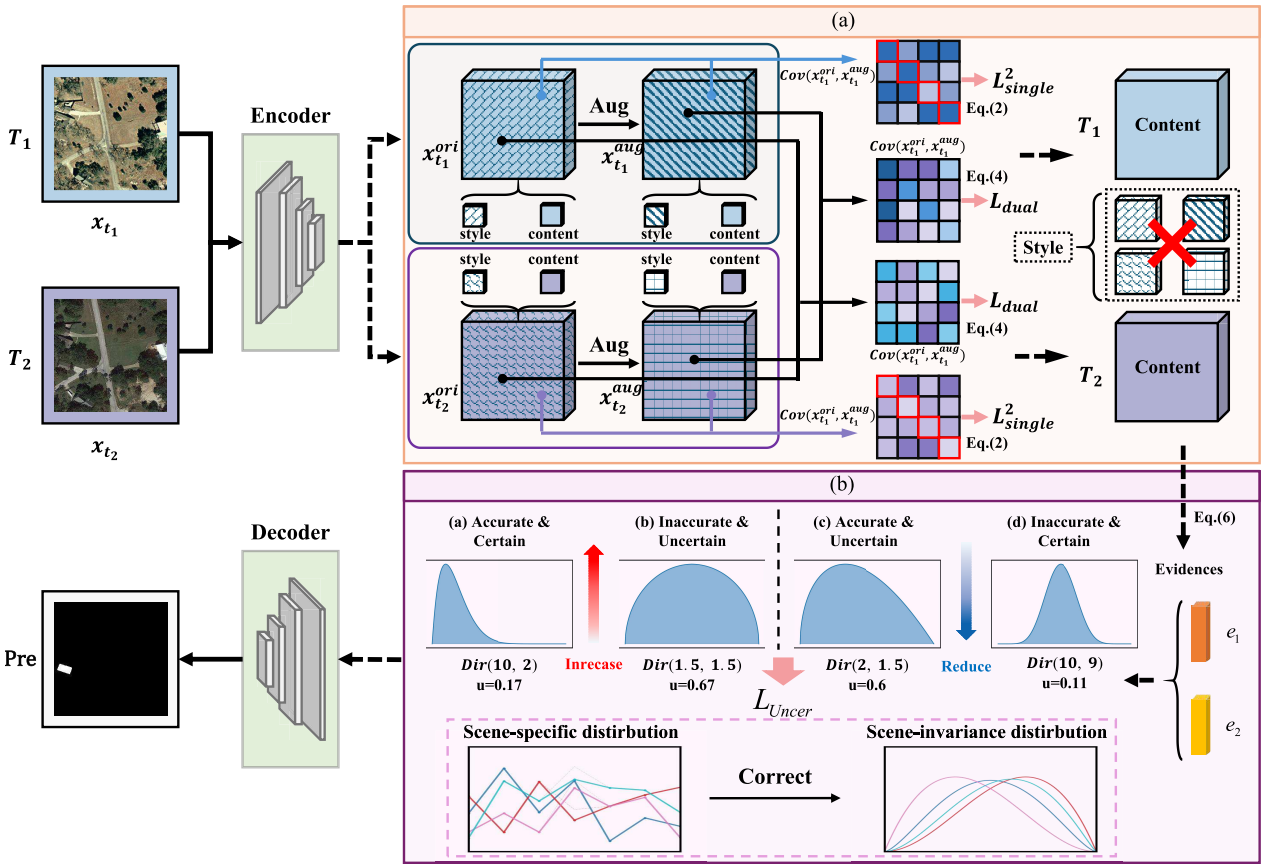


Fig. 1. Overview of the GenerChange method. The primary change detection backbone is used to obtain the basic features of the dual-temporal images, which contain both the style and content information. (a) Dual-covariance style-debias mechanism is proposed to remove the style information of the dual-temporal images. By carefully designing the dual-covariance style-debias mechanism, the style information of dual-temporal images is removed, and the content information is preserved, which reduces the aleatoric uncertainty. (b) It illustrates with an example of the RSCD task for ease of understanding the uncertainty estimation. Specifically, Dir(·) represents the probability distribution for change probabilities, and u denotes the uncertainty of the model. After training the model, it is expected that the model will exhibit higher uncertainty when the probability distributions are similar to change probabilities. In contrast, the uncertainty of the model is expected to be lower when the pixel change probabilities differ from the probability distributions. The evidence-aware uncertainty estimation module is applied to correct the distribution between seen scenes and unseen scenes. It effectively corrects the distribution between seen scenes and unseen scenes and reduces the epistemic uncertainty.

stereo estimation task. In recent years, many remote sensing image tasks have also introduced uncertainty estimation [60], [61], [62], [63] to quantify predictive errors and provide a basis for model optimization. In this work, we propose constraining the distribution of model predictions through uncertainty estimation in cross-domain change detection tasks, aiming to improve the model's cross-domain performance.

C. Domain Generalization

The objective of DG is to train models on multiple source domains and transfer them to any unseen target domain. Currently, DG methods [19], [20], [21], [22], [23], [24], [26], [64], [65], [66] have demonstrated excellent performance in basic classification tasks. However, when faced with complex tasks, more targeted DG designs are often required. For example, in recent years, there has been a surge of works in the field of autonomous driving semantic segmentation that integrates traditional DG strategies with the task at hand. Huang et al. [64] introduce iterative normalization (IterNorm) to enhance the efficiency of feature whitening. By iteratively aligning the eigenvalues, IterNorm optimizes the tradeoff between training stability and generalization. Zhang et al. [65] propose exact feature distribution matching (EFDM) to preserve content structure while transferring intricate texture patterns that conventional moment-matching fails to replicate. Zhang et al. [66] develop adversarial style augmentation (ASA) to explicitly model worst case style variations through adversarial statistical perturbations and then improve the generalization ability of the model. Ahn et al. [23] train the model to recognize features after style changes through covariance alignment and semantic space alignment, thereby mitigating the impact of feature whitening on semantic content. In this article, building on the traditional DG problem, we incorporate the concept of uncertainty estimation and tailor our approach for the task of change detection, achieving promising results.

III. Proposed Method

This section presents the pipeline of the GenerChange for DG RSCD, as illustrated in Fig. 1. The framework comprises two key components, including a dual-covariance style-debias mechanism and an evidence-aware uncertainty estimation module. First, we present the DG RSCD task formulation.

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The dual-covariance style-debias mechanism is designed to mitigate style differences by aligning dual-temporal covariances between original and augmented images. In addition, the evidence-aware uncertainty estimation module learns evidence for the Dirichlet distribution of pixel change probabilities, which corrects the distribution between the source domain and target domain. If the model exhibits low epistemic uncertainty, it maintains high reliability even when generalizing to domains with limited prior knowledge, thereby alleviating DG knowledge differences.

A. Formulations and Notations

We define $x_{t_1} \in \mathbb{R}^{H\times W\times 3}$ and $x_{t_2} \in \mathbb{R}^{H\times W\times 3}$ as input dual-temporal image pairs in the RSCD task, where H and W are the height and width of the image, x_{t_1} denotes the pretemporal image and x_{t_2} denotes the posttemporal image. In this study, the model is trained on the source domain and tested on the target domain. Specifically, the dataset source domain is defined as $S = \{(x_{t_1}^S, x_{t_2}^S, y^S)\}$, where $x_{t_1}^S, x_{t_2}^S \in \mathbb{R}^{H\times W\times 3}$ represent data from the source domain and $y^S \in \mathbb{R}^{H\times W\times 2}$ represents the corresponding binary label. The dataset of the target domain is defined as $T = \{(x_{t_1}^T, x_{t_2}^T, y^T)\}$, where $x_{t_1}^T, x_{t_2}^T \in \mathbb{R}^{H\times W\times 3}$ represents the data of the target domain and $y^T \in \mathbb{R}^{H\times W\times 2}$ is the corresponding binary label. Since our model is trained solely on source domain data, we adopt this notation for consistency in the following article and to streamline subsequent discussions. The superscript of $S = \{(x_{t_1}^S, x_{t_2}^S, y^S)\}$ is omitted in the partial symbol to simplify the symbol, and the simplified representation is denoted (x_{t_1}, x_{t_2}, y).

B. Dual-Covariance Style-Debias Mechanism

Existing DG approaches [23], [65], [66] focus primarily on single-temporal image detection tasks, neglecting the interference caused by style variations between dual-temporal images. This causes the model to struggle to distinguish semantic changes (e.g., building shape or location changes) from style variations. In addition, style variations randomly degrade semantic information, which introduces substantial aleatoric uncertainty. Aleatoric uncertainty originates from inherent irreducible randomness in data, mathematically modeled as the variance of observations. In dual-temporal RSCD, this randomness manifests as style differences, such as the shifts of season or illumination. Traditional data augmentation methods [20], [23], [26] further exacerbate this problem by disrupting the semantic correspondence between dual-temporal pairs. Therefore, there is a need for an approach to address the limitations brought by aleatoric uncertainty in DG RSCD tasks and to enhance the model's robustness against the style differences between dual-temporal images.

To address these challenges, we propose the dual-covariance style-debias mechanism. The core idea is to disentangle style and content through covariance alignment, where style discrepancies are suppressed, while semantic consistency is preserved. This aims to reduce the sensitivity of the model to style differences in dual-temporal images, thereby implicitly reducing the impact of aleatoric uncertainty on model performance and improving the model's generalization ability in target domains. The dual-covariance style-debias mechanism consists of two parts, including a single-temporal covariance module and a dual-temporal covariance module.

1) Single-Temporal Covariance Module: This module eliminates style discrepancies within the same temporal images. To remove the style, basically, we align the covariance matrices of the original and augmented images for the same temporal images, forcing the model to generate style-invariant features. To conveniently distinguish the style and content of images, we introduce covariance as a universal measurement. Specifically, we extract features using an RSCD backbone [14] and the features are defined as $x_{t_1}^{\text{ori}}, x_{t_2}^{\text{ori}}$. Then, they are augmented to generate features with a new style, which is defined as $x_{t_1}^{\text{aug}}, x_{t_2}^{\text{aug}}$. The covariance matrices for the same temporal original and augmented image features are calculated as $\text{Cov}(x_{t_i}^{\text{ori}}, x_{t_i}^{\text{ori}}) = (\text{IN}(x_{t_i}^{\text{ori}}))^T \cdot \text{IN}(x_{t_i}^{\text{ori}})$ and $\text{Cov}(x_{t_i}^{\text{aug}}, x_{t_i}^{\text{aug}}) = (\text{IN}(x_{t_i}^{\text{aug}}))^T \cdot \text{IN}(x_{t_i}^{\text{aug}})$. The $\text{IN}(\cdot)$ denotes the instance normalization [25]. The covariance matrices below are calculated using the same formula with different features. Then, L_{single}^1 is defined

$$L_{\text{single}}^1 = \sum_{i=1}^{2} \left\| \text{Cov}\left(x_{t_i}^{\text{ori}}, x_{t_i}^{\text{ori}}\right) - \text{Cov}\left(x_{t_i}^{\text{aug}}, x_{t_i}^{\text{aug}}\right) \right\|_2 \tag{1}$$

when $i = 1$, x_{t_i} denotes the pretemporal image and when $i = 2$, x_{t_i} denotes the posttemporal image.

In addition, to further suppress style-induced interference, we make feature independence between differently styled images that have the same content information. Specifically, if the diagonal elements of the covariance matrix $\text{Cov}(x_{t_i}^{\text{ori}}, x_{t_i}^{\text{aug}})$ are close to **1**, it implies that the style information of the original and augmented images is maximally independent (i.e., the style variations do not correlate with the content information). This independence ensures that the model's predictions remain consistent regardless of style differences. The covariance matrices between the original and augmented image features of the same temporal are calculated in the same way. Therefore, we design L_{single}^2 as follows:

$$L_{\text{single}}^2 = \sum_{i=1}^{2} \left\| 1 - \text{diag}\left(\text{Cov}\left(x_{t_i}^{\text{ori}}, x_{t_i}^{\text{aug}}\right)\right) \right\|_2 \tag{2}$$

where $\text{diag}(\text{Cov}(x_{t_i}^{\text{ori}}, x_{t_i}^{\text{aug}}))$ denotes the column vector comprising diagonal elements of $\text{Cov}(x_{t_i}^{\text{ori}}, x_{t_i}^{\text{aug}})$ and **1** denotes the one matrix. Furthermore, to achieve the better performance of removing the style information in dual-temporal images, we combine the L_{single}^1 and L_{single}^2 as follows:

$$L_{\text{single}} = L_{\text{single}}^1 + L_{\text{single}}^2. \tag{3}$$

The former loss function aims to remove the style information of every image and the latter loss function further considers the independence between style information and content information. Therefore, the style information of the dual-temporal images is disentangled and removed by combining these two loss functions.

2) Dual-Temporal Covariance Module: While style alignment reduces aleatoric uncertainty, excessive suppression can remove content information between dual-temporal images. To alleviate this, we design a dual-temporal covariance constraint

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to make the covariance matrix between the dual-temporal images not to tend toward one matrix. Therefore, the model preserves critical content information, enabling accurate separation of true semantic changes from style differences

$$L_{\text{dual}} = 1 - \left\| \text{diag}\left(\text{Cov}\left(x_{t_1}^{\text{ori}}, x_{t_2}^{\text{ori}}\right)\right) - \mathbf{1} \right\|_2 + 1 - \left\| \text{diag}\left(\text{Cov}\left(x_{t_1}^{\text{aug}}, x_{t_2}^{\text{aug}}\right)\right) - \mathbf{1} \right\|_2 \quad (4)$$

where $\text{diag}(\text{Cov}(x_{t_1}^{\text{ori}}, x_{t_2}^{\text{ori}})$ denotes the column vector comprising the diagonal elements of the covariance matrix between the original dual-temporal images, $\text{diag}(\text{Cov}(x_{t_1}^{\text{aug}}, x_{t_2}^{\text{aug}}))$ denotes the column vector comprising the diagonal elements of the covariance matrix between the augmented dual-temporal images, and $\mathbf{1}$ denotes the one matrix. Finally, we combine the loss function of single-temporal covariance and dual-temporal covariance to disentangle content information from style variations to reduce aleatoric uncertainty

$$L_{\text{Style}} = \lambda_1 \cdot L_{\text{single}} + \lambda_2 \cdot L_{\text{dual}} \quad (5)$$

where λ_1 and λ_2 means weighting hyperparameters of each loss functions of L_{Style}. By combining these two loss functions, our model removes the style information of dual-temporal images and preserves the content information. Therefore, the aleatoric uncertainty of the model is decreased and the model's robustness is enhanced.

C. Evidence-Aware Uncertainty Estimation Module

Existing RSCD models [14], [39], [40], [41], [42] predict probabilities of deterministic change, treating uncertainty as an implicit byproduct of limited training data. These approaches fail to quantify epistemic uncertainty and bridge knowledge gaps caused by characteristics of target domains. Inspired by EDL [37], which integrates subjective logic theory and Dempster–Shafer theory, this module addresses the fundamental limitation of conventional DG RSCD models, their inability to quantify epistemic uncertainty arising from domain gaps. Specifically, we propose an evidence-aware uncertainty estimation module that explicitly models and corrects pixel-wise change probabilities as a Dirichlet distribution, modeling uncertainty through an evidence aggregation process. In addition, the model with the correct pixel-wise change probability distribution can capture the correlation between dual-temporal images and knowledge about different domains. The model then focuses more on the domain-invariant information and ignores the domain-specific information. Therefore, this module enables uncertainty-aware predictions robust to domain shifts and improves the generalization ability of the model. The style-debias features obtained from the dual-covariance style-debias mechanism can be expressed as $x_{t_1}^{\text{debias}}$ and $x_{t_2}^{\text{debias}}$. Then, for a generic C-class classification, let $e \in R^C$ denote the evidence matrix predicted for style-debias features, the evidence for each class can be expressed as

$$e = f\left(x_{t_1}^{\text{debias}}, x_{t_2}^{\text{debias}}; \theta\right) = \left[e_1, \ldots, e_j, \ldots, e_C\right]. \quad (6)$$

Following EDL principles, a Dirichlet distribution is constructed $D(p \mid \alpha)$ parameterized by hyperparameters $\alpha = [\alpha_1, \ldots, \alpha_j, \ldots, \alpha_C]$, where $\alpha_j = e_j + 1$ to ensure nonnegativity. This parameterization transforms evidence into belief masses through

$$u + \sum_{j=1}^{C} b_j = 1, b_j = \frac{e_j}{S} \text{ and } u = \frac{C}{S} \quad (7)$$

where b_j represents the belief mass for class j, and u captures the total uncertainty due to limited evidence and $S = \sum_{j=1}^{C} \alpha_j$. However, standard evidential deep learning [37] outputs evidence vectors $\mathbf{e} = [e_1, \ldots, e_K]$ for single-temporal images with a K-class classification task. It fails to capture temporal dependencies in dynamic changes such as the land cover change caused by building construction. Our method combines the evidential deep learning and change detection that takes style-debiased feature pairs $(x_{t1}^{\text{debias}}, x_{t2}^{\text{debias}})$ as input and generates change detection evidence, including class of change (e_1) and class of unchange (e_2).

In addition, subjective logic formalizes the Dempster–Shafer theory of evidence, representing belief assignments over a frame of discernment as a Dirichlet distribution. This enables the quantification of belief masses and uncertainty using the principles of evidential theory within a well-defined framework. When $j = 1$, α_1 is related to evidence of the class of change, and when $j = 2$, α_2 is related to evidence of the class of unchange in the RSCD task. Furthermore, it is assumed that the probability mass $p = [p_1, \ldots, p_j, \ldots, p_C]$ follows a Dirichlet distribution parameterized by α. With the Dirichlet assumption, the expectation of the multinomial probability p is given by

$$E\left(p_j\right) = \frac{\alpha_j}{S} = \frac{\alpha_j}{\sum_{j=1}^{C} \alpha_j} = \frac{e_j + 1}{\sum_{j=1}^{C} e_i + C}. \quad (8)$$

In conventional softmax models, networks optimize by maximizing the point estimates of the correct class probabilities. This mechanism implicitly learns all the features that enhance prediction confidence, regardless of their cross-domain consistency. For example, domain-specific features such as terrain variations in target domains may be misinterpreted as building roof changes with spuriously high confidence. Unlike traditional models that output a scalar probability, our Dirichlet framework captures uncertainty, where high uncertainty indicates ambiguous predictions, while low uncertainty reflects confident predictions. This formulation enables modeling not just probabilities but also the uncertainty associated with them, effectively capturing second-order probabilities. In this context, Dirichlet evidence can be translated into subjective opinions using the following equations. The Dirichlet distribution $D(p \mid \alpha)$ is parameterized by α

$$D\left(p \mid \alpha\right) = \begin{cases} \dfrac{1}{B\left(\alpha\right)} \displaystyle\prod_{j=1}^{C} p_j^{\alpha_j - 1}, & \text{for } p \in S_C \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

where S_C is the C-dimensional unit simplex and $B(\alpha)$ is the C -dimensional multinomial beta function.

As shown in Fig. 1, the Dirichlet distribution on a 2-D simplex can be viewed as a Beta distribution curve, and the two ends of the curve represent two point estimates of the probability distribution, also with a range of [0, 1]. Here,

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$D(p|\alpha)$ is considered as the prior of the likelihood function $L(y|p)$, and the marginal likelihood function can be derived as

$$M(y \mid \alpha) = \int_{p \in S_C} L(y \mid p) D(p \mid \alpha) \, \mathrm{dp} = \int_{p \in S_C} \prod_{j=1}^{C} p_j^{y_j} \frac{1}{B(\alpha)} \prod_{j=1}^{C} p_j^{\alpha_j - 1} \, \mathrm{dp} \tag{10}$$

where y is the label of the data. Therefore, using the negative logarithm of the marginal likelihood function as the final optimization goal of the model can be expressed as

$$L_{\mathrm{Uncer}}(y, e; \theta) = \sum_{i \in \{c,u\}} y \left(\log S - \log (e_c + 1) \right). \tag{11}$$

This loss function drives the model to balance evidence accumulation with uncertainty reduction. By explicitly quantifying epistemic uncertainty, the module corrects the distribution between seen and target domains. Furthermore, the corrected distribution makes the model focus more on the domain-invariant information in the source domain, ensuring reliable DG.

The total loss function of our SingNet is defined as

$$L_{\mathrm{Sing}} = \omega_1 \cdot L_{\mathrm{Style}} + \omega_2 \cdot L_{\mathrm{Uncer}} \tag{12}$$

where ω_1 and ω_2 denote the weighting factor of each loss functions. As these two loss functions are combined, our model removes the style information of the dual-temporal images and reduces the aleatoric uncertainty. Then, the evidence is gathered from features, and a Dirichlet distribution is built to correct the distribution of change probability and capture correlation about different domains. Finally, the model focuses on domain-invariant information, and the epistemic uncertainty is reduced.

IV. Experiments

In this section, we detail the implementation, experimental setup for comparison with existing change detection methods and general DG approaches, as well as ablation studies conducted to further substantiate the effectiveness of our method.

A. Datasets

We validate our model using four RSCD datasets, all of which contain labels specific to building changes.

1) LEVIR-CD [10]: The LEVIR-CD dataset is a large-scale benchmark for building change detection, constructed from 637 pairs of very high-resolution (0.5 m/pixel) Google Earth images with 1024 × 1024 pixel size, spanning 5–14 years. These bitemporal patches capture urban development changes (e.g., building construction/demolition) in 20 regions in Texas, USA, including Austin, Lakeway, and Buda. Expert-annotated binary labels (1 for changed, 0 for unchanged) ensure high-quality ground truth, covering 31 333 building change instances. The dataset's geographic and temporal diversity (2002–2018 acquisition dates) introduces natural variations in illumination and seasonal conditions, supporting robust model training for real-world scenarios.

2) WHU-CD [67]: The WHU-CD dataset is a high-resolution aerial imagery benchmark for postearthquake building change detection comprising two large-scale aerial images (32 507 × 15 354 pixels each) captured in 2012 and 2016 with a spatial resolution of 0.075 m. Focused on Christchurch, New Zealand, it documents urban changes following the 2010–2011 earthquakes, including building damage, demolition, and reconstruction. The very high resolution of the dataset enables the precise identification of structural alterations at the individual building level.

3) EGY-BCD [68]: The EGY-BCD dataset is a benchmark for building change detection in urban development, comprising bitemporal very high-resolution (0.25 m/pixel) satellite images (256 × 256 pixels) captured between 2017 and 2022. It covers four regions in Egypt: New Mansoura, El Galala City, New Cairo, and New Thebes. The dataset exhibits natural variations in lighting conditions across acquisition periods, aiding models in distinguishing relevant changes from transient environmental effects. Designed for robust evaluation, EGY-BCD supports applications in urban planning and land management through its geographically diverse and temporally rich imagery.

4) GZ-CD [69]: The GZ-CD dataset is a suburban change detection benchmark comprising 19 very high-resolution (0.55 m/pixel) Google Earth image pairs collected between 2006 and 2019, covering Guangzhou, China. These RGB images, with sizes ranging from 1006 × 1168 to 4936 × 5224 pixels, capture seasonal variations and suburban urbanization changes (e.g., land-use transitions and infrastructure development). The dataset's multiyear span and spatial diversity (across Guangzhou's suburban zones) enhance model robustness against environmental fluctuations. Designed for monitoring urban expansion, GZ-CD supports large-scale and fine-grained analysis of suburban dynamics through its varied resolutions and temporal depth.

B. Compared Methods

We compare our method with other DG approaches: the baseline (BIT-CD [14] trained on the source domain), IterNorm [64], CrossNorm [70], EFDM [65], Contra-ACE [71], ADVstyle [66], Blind-Net [23], and PerXAN [72].

1) BIT-CD [14]: The proposed method introduces a dual-temporal image transformer (BIT) to model contextual dependencies in bi-temporal images. Instead of dense pixel-wise correlation, BIT represents images using semantic tokens that capture high-level change concepts (e.g., buildings and vegetation). These tokens are processed in a token-based spatiotemporal space to efficiently model global context. In addition, pixel-token affinity matrices are designed to enhance pixel-level features by aggregating token-guided semantics.

2) IterNorm [64]: This work introduces IterNorm to enhance feature whitening efficiency over batchnorm (BN) and decorrelated BN (DBN). While BN standardizes activations without decorrelation and DBN achieves whitening via computationally heavy eigen-decomposition, IterNorm employs Newton's iterations to approximate whitening matrices, eliminating eigen-decomposition while maintaining GPU efficiency. In addition, stochastic normalization disturbance (SND) is

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proposed to quantify mini-batch sampling uncertainty, explaining DBN's group-wise superiority and BN's batch-size sensitivity. By aligning eigenvalues, IterNorm optimizes the tradeoff between stability and generalization.

3) CrossNorm [70]: This work introduces crossnorm and selfnorm, dual normalization mechanisms to address distribution shifts through channel-wise statistics manipulation. Crossnorm augments training diversity by swapping feature map statistics (mean/variance) across samples, which simulates style variations while preserving semantic content. Selfnorm complements this by employing attention-driven recalibration to align test-time feature statistics with training distributions, bridging domain gaps. By synergizing style expansion and dynamic adaptation, the method achieves robust generalization.

4) EFDM [65]: EFDM is proposed to bypass distributional assumptions by directly matching empirical cumulative distribution functions (eCDFs) of features. This technique leverages exact histogram matching (EHM) in the feature space—a plug-and-play module that aligns eCDFs through a novel sort-matching algorithm. For style transfer, EFDM preserves content structure while transferring intricate texture patterns that conventional moment-matching fails to replicate. In DG, EFDM reduces domain gaps through its alignment ability.

5) Contra-ACE [71]: In the realm of DG, this study proposes a Contrastive-ACE learning framework. This framework comprises a dual-branch architecture designed to explicitly model causal invariance through average causal effect (ACE) estimation, where cross-domain samples of the same class form positive pairs and different classes form negative pairs. In addition, to mitigate spurious correlations induced by domain-specific biases, a dynamic feature weighting strategy is introduced, prioritizing causal features over style-related attributes. Furthermore, to disentangle domain-invariant causal mechanisms from domain-specific variations, a mutual information constraint is formulated. This constraint leverages intraclass ACE consistency across domains and interclass ACE divergence, ensuring robust generalization to unseen target domains.

6) ADVstyle [66]: For DG under style shifts, this research introduces ASA to explicitly model worst case style variations through adversarial statistics perturbations. ASA integrates a dual-phase architecture that first generates style perturbations by maximizing task loss (simulating extreme domain shifts) and then adapts models via gradient reversal layers (GRLs). Furthermore, to balance perturbation diversity and semantic invariance, an adversarial regularization term is formulated through cross-domain contrastive constraints—minimizing intraclass style discrepancy while maximizing interclass separability.

7) Blind-Net [23]: For domain-generalized semantic segmentation, this method introduces Blind-Net, an encoder–decoder framework that mitigates style bias while preserving content integrity. BlindNet employs covariance alignment to normalize style variations in the encoder, enabling uniform feature representation across domains without discarding critical content cues. Furthermore, a semantic consistency contrastive learning mechanism is integrated into the decoder to disentangle class-discriminative features and suppress style-induced misclassifications, which can hardly be used for the DG RSCD task.

8) PerXAN [72]: For federated DG, this method introduces PerXAN, a personalized normalization architecture that disentangles domain-specific biases via explicitly assembled normalization (PXAN). PXAN integrates global instance normalization to filter client-specific style features while retaining local batch normalization for discriminative semantics, ensuring privacy preservation without cross-client data sharing. Furthermore, to bridge the gap between domain-invariant representation learning and federated aggregation, a geometric-aware regularizer is formulated. This regularizer aligns client models' feature distributions with the global classifier through contrastive invariance constraints, preventing over-suppression of discriminative features.

9) SAAChange [73]: It employs a selective adversarial adaptation framework. The framework utilizes two domain discriminators. The first identifies individual source domains to select a crucial subset of source samples for training, thereby enhancing positive transfer and mitigating negative transfer. Concurrently, the second discriminator aligns the distribution of the selected source samples with the target domain via adversarial learning. Furthermore, to adapt the model to the unlabeled target domain, reliable pseudo-labels are generated using homogeneity- and boundary-based strategies. These pseudo-labels, alongside the selected source data, are then used to jointly fine-tune the change detection model.

10) SSLChange [74]: It explores a novel self-supervised contrastive framework applicable to RSCD. SSLChange accomplishes self-learning using only single-temporal samples and can be flexibly transferred to mainstream change detection baseline models. Through self-supervised contrastive learning, feature representation pretraining is performed directly on raw data without requiring any labeling. Subsequently, with limited labeled data, the pretrained features are aligned with annotations via fully supervised fine-tuning. Without introducing additional data or labels, significant performance enhancement is achieved for downstream baselines.

C. Implementation Details

The experimental environment is based on the PyTorch framework, the learning rate is set to 0.01, and using a stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of $5e^{-4}$. The training process involves 100 epochs with a batch size of 4. A polynomial learning rate schedule is adopted, starting with an initial learning rate of 0.01, which is linearly decayed to 0 by the 100th epoch. We utilize the BIT-CD [14] model with cross-entropy loss as the primary architectural framework for change detection. The model uses ResNet-18 and ResNet-50 as the backbone and the augmented data is generated through [20]. This model is implemented using the PyTorch framework and is trained on a NVIDIA RTX 3090 GPU with 24 GB of video memory. The images in every dataset are processed into a format of 256 × 256. The weighting parameters of ω_1 and ω_2 of $\mathcal{L}_{\text{Sing}}$ are set at 0.2 and the weighting parameters of λ_1 and λ_2 of $\mathcal{L}_{\text{Style}}$ are set at 1, for convenient comparison. And the tradeoff of these

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TABLE I

QUANTITATIVE COMPARISON OF ACC (%) ACROSS STATE-OF-THE-ART METHODS. BEST RESULTS ARE SHOWN IN BOLD. SECOND-BEST ARE SHOWN IN BLUE

Backbone	Training	Testing	IterNorm[64] CVPR2019 Acc (%)	CrossNorm[70] ICCV2021 Acc (%)	EFDM[65] CVPR2022 Acc (%)	Contr-ACE[71] TIP2022 Acc (%)	ADVstyle[66] AAAI2023 Acc (%)	Blind-net[23] CVPR2024 Acc (%)	PerXAN[72] CVPR2024 Acc (%)	Ours Acc (%)
BIT-ResNet-50	LEVIR-CD	WHU-CD	95.74	95.79	96.35	**96.76**	96.64	95.38	96.46	**97.63**
		GZ-CD	91.86	92.20	92.38	91.08	92.64	**93.02**	**93.29**	92.98
		EGY-BCD	93.66	92.29	93.14	92.86	93.13	**93.79**	92.71	**94.71**
		Avg	93.75	93.42	93.96	93.57	94.14	94.06	**94.15**	**95.11**
		Improv.	+1.36	+1.69	+1.15	+1.54	+0.97	+1.05	+0.96	-
	WHU-CD	LEVIR-CD	94.58	**95.06**	94.78	94.98	94.76	93.71	94.74	**95.22**
		GZ-CD	89.91	89.31	90.66	88.14	**91.16**	91.04	88.87	**92.32**
		EGY-BCD	88.90	91.32	91.72	90.38	85.99	**92.97**	85.47	**92.77**
		Avg	91.13	91.90	92.39	91.17	90.64	**92.57**	89.69	**93.44**
		Improv.	+2.31	+1.54	+1.05	+2.27	+2.80	+0.87	+3.75	-
BIT-ResNet-18	LEVIR-CD	WHU-CD	96.26	96.21	**96.95**	**96.64**	96.62	95.37	95.98	94.76
		GZ-CD	92.09	92.47	91.97	**93.29**	**92.83**	92.60	91.87	92.68
		EGY-BCD	**93.47**	**93.44**	92.65	93.35	93.27	92.54	92.94	93.38
		Avg	93.94	94.04	93.85	**94.43**	**94.24**	93.50	93.60	93.61
		Improv.	-0.33	-0.43	-0.24	-0.82	-0.63	+0.11	+0.01	-
	WHU-CD	LEVIR-CD	94.68	94.79	91.77	**95.95**	94.89	95.06	94.89	**95.24**
		GZ-CD	87.30	87.15	89.54	**91.84**	88.95	87.59	90.63	**92.16**
		EGY-BCD	91.74	91.00	84.69	89.19	92.33	**93.30**	90.01	**93.64**
		Avg	91.24	90.98	88.67	**92.33**	92.06	91.98	91.84	**93.68**
		Improv.	+2.44	+2.70	+5.01	+1.35	+1.62	+1.70	+1.84	-

TABLE II

QUANTITATIVE COMPARISON OF MIOU (%) ACROSS STATE-OF-THE-ART METHODS. BEST RESULTS ARE SHOWN IN BOLD. SECOND-BEST ARE SHOWN IN BLUE

Backbone	Training	Testing	IterNorm[64] CVPR2019 mIoU (%)	CrossNorm[70] ICCV2021 mIoU (%)	EFDM[65] CVPR2022 mIoU (%)	Contr-ACE[71] TIP2022 mIoU (%)	ADVstyle[66] AAAI2023 mIoU (%)	Blind-net[23] CVPR2024 mIoU (%)	PerXAN[72] CVPR2024 mIoU (%)	Ours mIoU (%)
BIT-ResNet-50	LEVIR-CD	WHU-CD	70.63	70.55	70.92	**73.83**	73.16	69.42	73.73	**77.22**
		GZ-CD	59.53	64.05	62.10	55.62	62.43	**67.72**	**66.83**	63.87
		EGY-BCD	58.29	59.06	59.75	52.24	53.46	**61.63**	58.90	**64.81**
		Avg	62.82	64.55	64.26	60.56	63.01	66.26	**66.49**	**68.63**
		Improv.	+5.81	+4.08	+4.37	+8.07	+5.62	+2.37	+2.14	-
	WHU-CD	LEVIR-CD	50.65	50.92	50.67	50.63	50.75	**56.02**	52.53	**54.02**
		GZ-CD	55.42	57.21	59.44	53.26	59.66	**61.84**	60.42	**66.16**
		EGY-BCD	51.38	48.46	53.19	48.12	46.93	**55.72**	48.14	**54.81**
		Avg	52.48	52.20	54.43	50.67	52.45	**57.86**	53.69	**58.33**
		Improv.	+5.85	+6.13	+3.90	+7.66	+5.88	+0.47	+4.64	-
BIT-ResNet-18	LEVIR-CD	WHU-CD	71.78	71.60	**73.31**	71.26	**72.30**	68.38	71.10	68.04
		GZ-CD	61.44	61.55	59.35	62.52	63.23	**65.72**	59.95	**69.45**
		EGY-BCD	54.11	55.07	53.28	62.70	54.68	61.09	**62.82**	**63.71**
		Avg	62.45	62.74	61.98	**65.50**	63.41	65.07	64.62	**67.07**
		Improv.	+4.62	+4.33	+5.09	+1.57	+3.66	+2.00	+2.45	-
	WHU-CD	LEVIR-CD	50.05	51.14	48.73	51.35	50.15	**51.65**	50.77	**53.97**
		GZ-CD	52.55	54.35	59.26	**60.00**	54.71	53.13	56.96	**63.21**
		EGY-BCD	52.08	51.25	46.03	49.74	51.82	**55.16**	47.72	**56.78**
		Avg	51.56	52.25	51.34	**53.70**	52.23	53.31	51.82	**57.98**
		Improv.	+6.42	+5.73	+6.64	+4.28	+5.75	+4.67	+6.16	-

parameters is discussed in the ablation study. To evaluate DG RSCD model performance, the mean intersection over union (mIoU) metric is considered for quantitative research.

D. Performance Comparison

To evaluate the DG ability of our model, we conduct experiments under two scenarios: 1) training on LEVIR-CD and testing on WHU-CD, EGY-BCD, and GZ-CD; and 2) training on WHU-CD and testing on LEVIR-CD, EGY-BCD, and GZ-CD. Specifically, we separately compare our method with normal DG methods and domain adaptation change detection methods. Table I presents the Acc comparison across different backbone architectures. Table II shows the mIoU comparison, and Table III reports the mean $F1$ (mF1) results, all evaluated under both training sets.

For the model with a BIT-based ResNet-50 backbone, when trained in LEVIR-CD, existing DG methods exhibit limited generalization capability. For example, IterNorm [64] achieves an average 62.82% mIoU and 72.21% mF1 by whitening the feature style, while CrossNorm [70] improves this to 64.55% and 74.41% by cross-domain normalization. ADVstyle [66] further improves performance to 63.01% average mIoU by generating stylized samples, and PerXAN [72] obtains the suboptimal performance through style-invariant feature preservation. When trained on the WHU-CD dataset, our method achieves 58.33% average mIoU, surpassing the Blind-Net [23] method by 0.47%. When trained on the LEVIR-CD dataset using BIT-based ResNet-18 backbone, our method achieves 93.61% average Acc, 67.07% mIoU, and 77.17% mF1, representing 0.01%, 2.45%, and 2.71% improvement over the PerXAN [72] method. This is because the proposed method can remove the dual-temporal images and reduce the aleatoric uncertainty. Furthermore, our designed evidence-aware uncertainty estimation module is utilized to predict a Dirichlet distribution of class probabilities. It can be viewed as a process of gathering evidence which can be used to correct the different distributions and gather the scene-invariant information. When the model exhibits low uncertainty, it

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TABLE III

QUANTITATIVE COMPARISON OF MEAN $F1$ (m$F1$, %) ACROSS STATE-OF-THE-ART METHODS. BEST RESULTS ARE SHOWN IN BOLD. SECOND-BEST ARE SHOWN IN BLUE

Backbone	Training	Testing	IterNorm[64] CVPR2019 mF1 (%)	CrossNorm[70] ICCV2021 mF1 (%)	EFDM[65] CVPR2022 mF1 (%)	Contr-ACE[71] TIP2022 mF1 (%)	ADVstyle[66] AAAI2023 mF1 (%)	Blind-net[23] CVPR2024 mF1 (%)	PerXAN[72] CVPR2024 mF1 (%)	Ours mF1 (%)
BIT-ResNet-50	LEVIR-CD	WHU-CD	80.23	80.14	80.37	**82.93**	82.35	79.15	82.90	**85.64**
		GZ-CD	69.36	74.49	72.24	64.52	72.54	**78.06**	**77.10**	74.05
		EGY-BCD	67.06	68.60	69.14	58.59	60.38	**71.21**	68.24	**74.56**
		Avg	72.21	74.41	73.92	68.68	71.76	**76.14**	76.08	**78.08**
		Improv.	+5.87	+3.67	+4.16	+9.40	+6.32	+1.94	+2.00	-
	WHU-CD	LEVIR-CD	54.91	55.09	54.83	54.63	54.98	**64.25**	58.01	**60.15**
		GZ-CD	64.76	67.39	69.68	62.51	69.77	71.38	**71.46**	**76.67**
		EGY-BCD	59.34	53.07	60.68	53.05	53.62	**64.59**	55.97	**62.60**
		Avg	59.67	58.52	61.73	56.73	59.46	**66.74**	61.82	**66.47**
		Improv.	+6.80	+7.95	+4.74	+9.74	+7.01	-0.27	+4.65	-
BIT-ResNet-18	LEVIR-CD	WHU-CD	81.19	81.03	**82.43**	80.62	**81.59**	78.13	80.62	77.95
		GZ-CD	71.58	71.58	69.09	72.44	73.39	**76.15**	69.87	**79.81**
		EGY-BCD	61.21	62.66	60.35	72.62	62.16	71.01	**72.88**	**73.75**
		Avg	71.32	71.75	70.62	**75.22**	72.38	75.09	74.45	**77.17**
		Improv.	+5.84	+5.42	+6.55	+1.95	+4.79	+2.08	+2.72	-
	WHU-CD	LEVIR-CD	53.78	55.64	53.27	55.30	53.83	**56.35**	54.93	**60.07**
		GZ-CD	61.88	64.49	69.86	**69.93**	64.20	62.58	66.54	**73.57**
		EGY-BCD	58.94	58.01	52.84	56.55	58.21	**62.85**	52.56	**65.01**
		Avg	58.20	59.38	58.66	**60.60**	58.74	**60.60**	58.01	**66.22**
		Improv.	+8.02	+6.84	+7.56	+5.62	+7.48	+5.62	+8.21	-

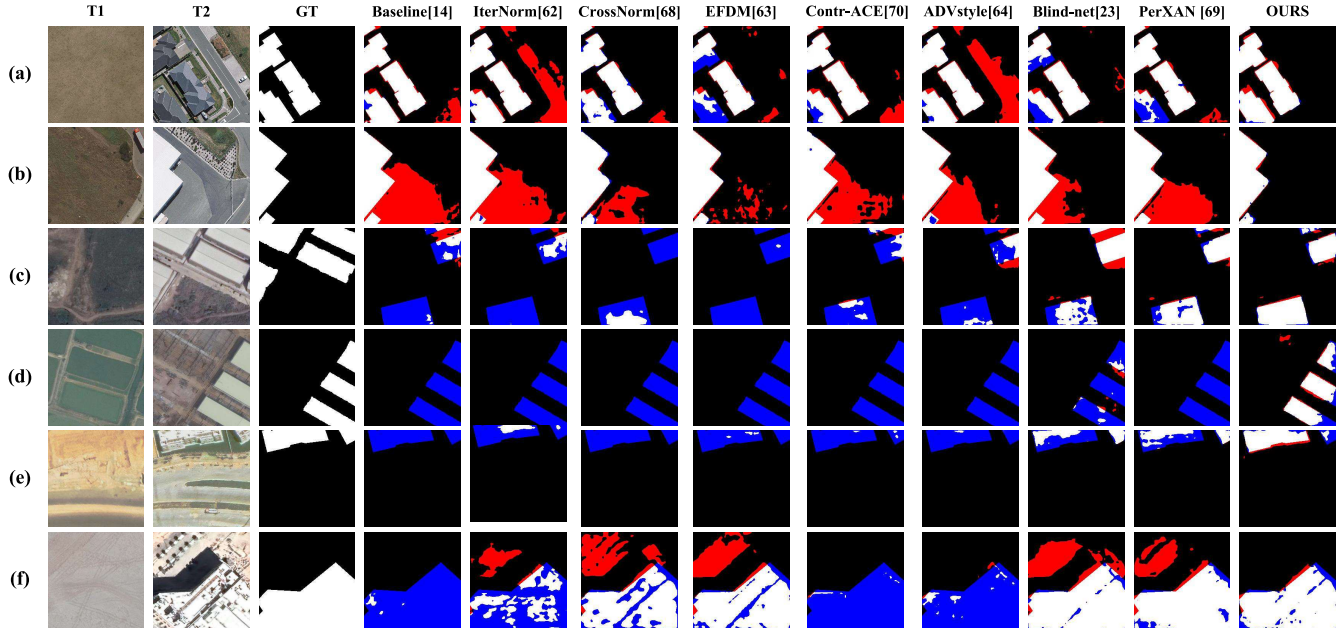


Fig. 2. Visualization of cross-domain change detection performance for methods trained on the LEVIR-CD dataset (ResNet-50 backbone architecture) and evaluated on three distinct test sets: WHU-CD, GZ-CD, and EGY-BCD. (a) and (b) Demonstrate model predictions on WHU-CD, (c) and (d) demonstrate model predictions on GZ-CD, and (e) and (f) demonstrate model predictions on EGY-BCD datasets, respectively, highlighting generalization capabilities under domain shifts. Correctly predicted changed areas are denoted as TPs (white). Incorrectly predicted changed areas where no actual change exists are denoted as FPs (red). Actual changed areas missed by the model are denoted as FNs (blue).

mitigates the issue caused by the lack of knowledge in unseen scenes. Such consistent advancements across Acc, mIoU, and mF1 metrics underscore the framework's robustness against domain shifts, particularly in challenging scenarios involving seasonal variations.

Fig. 2 visually confirms these quantitative improvements, showing improved consistency in change detection maps across heterogeneous scenes. In the WHU-CD dataset, baseline predictions generate extensive false positives (FPs), shown as red patches, between land and building variations misclassified as structural changes. Our method eliminates most of these errors. Crucially, in the GZ-CD dataset, the similarity in color between land and buildings triggers false negatives (FNs), shown as blue omissions in the baseline results. In the EGY-BCD dataset, dust and sand encroachment cause competing methods to either over-detect desertification patterns as changes or miss actual urban constructions under dust. By correlating epistemic uncertainty with domain-shifted regions, our method reduces FPs while simultaneously recovering dust-occluded structures by contiguous white true positive (TP) blocks, maintaining structural integrity where others fragment. The visualizations reveal our method's capability to maintain

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TABLE IV
COMPUTATIONAL COST COMPARISON WITH A RESNET-50 BACKBONE ON A NVIDIA 3090 GPU WITH AN IMAGE RESOLUTION OF 256 × 256. OUR RESULTS ARE SHOWN IN BOLD

Models	Params (M)	GFLOPs	TIME (s)
Baseline [14]	26.49	199.50	0.0514
Iterative Norm [64]	26.49	199.44	0.0602
CrossNorm [70]	26.55	84.38	0.0524
EFDM [65]	26.49	199.50	0.0537
Contra-ACE [71]	26.49	199.48	0.0487
ADVstyle [66]	24.45	199.50	0.0489
Blind-net [23]	26.50	199.50	0.0494
PerXAN [72]	26.49	199.57	0.0553
OURS	**26.50**	**199.50**	**0.0541**

the structural integrity of changed objects under significant style variations while suppressing FPs caused by style differences. It shows that our framework demonstrates superior performance in DG RSCD, particularly through content information preservation.

In addition, to confirm that our method does not introduce additional computational inference cost, we evaluate the parameter counts, GFLOPS, and inference time as shown in Table IV. Parameter counts of our method remain nearly identical at 26.50 M versus the baseline 26.49 M, and computational costs maintain the same at 199.50 GFLOPs. Furthermore, inference latency increases negligibly from 0.0514 to 0.0541 s/image. The observed training overhead thus constitutes a strategically justified investment, trading temporary computational expenditure for significantly enhanced cross-DG while maintaining operational efficiency in practical applications. Although our method does not have an advantage over other methods, it is enough for our method to satisfy the normal requirement. This ensures that the model's real-time applicability in target domains remains practical and efficient.

To compare the performance of our method and the domain adaptation change detection methods, we separately train these methods on the LEVIR-CD dataset and the WHU-CD dataset with different backbones. Compared to DG methods, the SAAChange method and the SSLChange method demonstrate distinct domain adaptation characteristics when incorporating limited unlabeled target-domain data during training, as evidenced in Table V. Unlike DG methods that operate without any target-domain information, these approaches leverage minimal unsupervised target information. Across all configurations, our method achieves the highest average performance. Specifically, our method achieves 68.63% average mIoU with the BIT-ResNet-50 backbone and 67.07% with the BIT-ResNet-18 backbone, when training the model on the LEVIR-CD dataset. These results surpass SAAChange by 2.05% and 1.88%, and outperform SSLChange by 1.41% and 1.99%. We analyze the WHU-CD dataset, which is characterized by high-resolution aerial images of postdisaster urban environments with complex structural details and variable shadow patterns. Our method, trained on the LEVIR-CD dataset with BIT-ResNet-50 backbone, achieves 77.22% mIoU, outperforming the SAAChange method by 0.08% and the SSLChange method by 1.36%. In addition, we analyze the GZ-CD dataset, which features long-term suburban-urban transitions with pronounced seasonal variations. Our method with BIT-ResNet-18 backbone achieves 69.45% mIoU, exceeding the SAAChange method by 5.30% and the SSLChange method by 2.35%. The EGY-BCD dataset presents desert conditions with sand-reflected glare that amplify both aleatoric and epistemic uncertainties. Here, our method trained on the LEVIR-CD dataset with the BIT-ResNet-50 backbone reaches 64.81% mIoU, surpassing the SSLChange method by 3.61% and the SAAChange method by 6.02%.

E. Ablation Experiments

In this section, we conduct a series of ablation studies to demonstrate the individual contributions and effectiveness of the components of our GenerChange. Using the BIT-CD model with a ResNet-50 backbone, we train on LEVIR-CD and test on EGY-BCD and GZ-CD. Table VI illustrates the impact of the two modules on DG performance. Specifically, the baseline model using only the dual-covariance style-debias mechanism and the model using only the evidence-aware uncertainty estimation module exhibit suboptimal performance on the unseen scenes. In contrast, the combined application of both modules, complementing each other, yields better results. More specifically, the aleatoric uncertainty style-debias optimization module is effective in removing data aleatoric uncertainty but lacks optimization for the model's overall generalization performance. The epistemic uncertainty optimization module is crucial for correcting the distribution and scene-invariant information extraction, enabling the construction of a more robust predictive distribution. However, without style-debias, style differences negatively impact the reliability of evidence.

1) Effectiveness of Dual-Covariance Style-Debias Mechanism: As shown in Table VI, we investigate the contributions of the proposed dual-covariance style-debias mechanism. We conduct experiments by removing the evidence-aware uncertainty estimation module, denoted as L_{Uncer} in the table, with all loss functions crossed indicating the baseline. The mechanism comprises three components: L^1_{single}, L^2_{single}, and L_{dual}. The results of the LEVIR-CD dataset demonstrate that using only L^1_{single} improves the Acc, mIoU, and mF1 by 1.58%, 3.71%, and 1.91% over baseline. This validates L^1_{single}'s role in preliminarily eliminating dual-temporal image style differences. When adding L^2_{single}, performance further improves by 0.25%, 0.33%, and 0.03%, showing its capacity to enhance feature independence between stylistically different. Finally, incorporating L_{dual} increases results by 0.17%, 1.62%, and 2.05%, confirming the necessity to preserve dual-temporal content consistency for separating true semantic changes from stylistic variations.

2) Effectiveness of Evidence-Aware Uncertainty Estimation Module: We have researched the effectiveness of the EDL layer with the Dirichlet distribution of the neural network, as shown in Table VI. Concretely, to estimate the effectiveness of this module, we removed the dual-covariance style-debias mechanism. Compared to baseline, Acc and mIoU improve by 1.64% and 1.88% while mF1 decreases by 0.71%, indicating the module's inherent capacity to reduce epistemic uncertainty

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TABLE V

QUANTITATIVE COMPARISON OF ACC, MIOU, AND MEAN $F1$ ($mF1$) (%) ACROSS STATE-OF-THE-ART METHODS. BEST RESULTS ARE SHOWN IN BOLD, SECOND-BEST IN BLUE

Backbone	Training	Testing	SAAChange[73] TGRS2020			SSLChange[74] TGRS2024			Ours		
			Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)
BIT-ResNet-50	LEVIR-CD	WHU-CD	**97.46**	**77.14**	**85.60**	97.19	75.86	84.59	**97.63**	**77.22**	**85.64**
		GZ-CD	**92.90**	63.82	74.03	92.83	**64.61**	**74.90**	**92.98**	**63.87**	**74.05**
		EGY-BCD	**93.88**	58.79	67.62	93.74	**61.20**	**70.72**	**94.71**	**64.81**	**74.56**
		Avg	**94.75**	66.58	75.75	94.59	**67.22**	**76.74**	**95.11**	**68.63**	**78.08**
		Improv.	+0.36	+2.05	+2.33	+0.52	+1.41	+1.35		-	
	WHU-CD	LEVIR-CD	**95.00**	50.66	54.67	94.96	**55.59**	**62.68**	**95.22**	**54.02**	**60.15**
		GZ-CD	**91.91**	**64.49**	**75.04**	90.43	61.08	71.71	**92.32**	**66.16**	**76.67**
		EGY-BCD	**89.67**	**51.06**	**58.42**	84.77	46.87	54.24	**92.77**	**54.81**	**62.60**
		Avg	**92.19**	**55.40**	62.71	90.05	54.51	**62.88**	**93.44**	**58.33**	**66.47**
		Improv.	+1.25	+2.93	+3.76	+3.39	+3.82	+3.60		-	
BIT-ResNet-18	LEVIR-CD	WHU-CD	**96.78**	**73.70**	**82.81**	**96.07**	**70.74**	**80.26**	94.76	68.04	77.95
		GZ-CD	92.13	64.15	74.61	**92.89**	**67.10**	**77.47**	**92.68**	**69.45**	**79.81**
		EGY-BCD	93.23	57.41	66.05	**93.41**	**57.74**	**66.43**	**93.38**	**63.71**	**73.75**
		Avg	**94.12**	**65.19**	**74.72**	**94.05**	65.08	74.49	93.61	**67.07**	**77.17**
		Improv.	-0.51	+1.88	+2.45	-0.44	+1.99	+2.68		-	
	WHU-CD	LEVIR-CD	94.32	**55.51**	**62.88**	**94.73**	**57.89**	**66.06**	**95.24**	53.97	60.07
		GZ-CD	89.70	55.05	64.34	**91.07**	**57.57**	**67.16**	**92.16**	**63.21**	**73.57**
		EGY-BCD	90.94	**55.57**	**64.52**	**91.21**	53.67	61.66	**93.64**	**56.78**	**65.01**
		Avg	91.65	55.38	63.91	**92.34**	**56.38**	**64.96**	**93.68**	**57.98**	**66.22**
		Improv.	+2.03	+2.61	+2.31	+1.34	+1.61	+1.26		-	

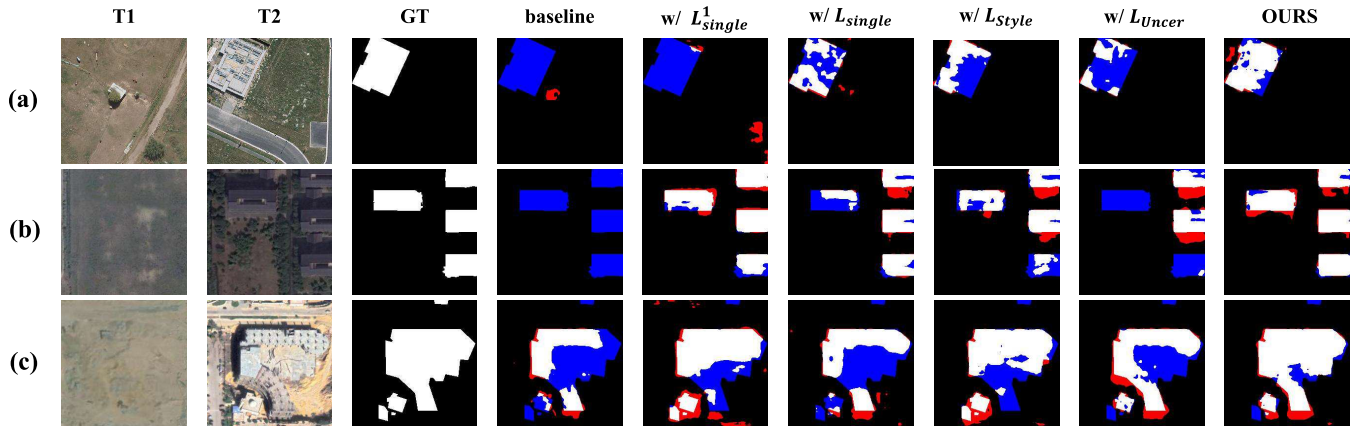


Fig. 3. Visualization of cross-domain change detection performance for methods trained on the LEVIR-CD dataset with BIT-ResNet-50 backbone architecture and evaluated on three test sets: WHU-CD dataset, GZ-CD dataset, and EGY-BCD dataset. Correctly predicted changed areas are denoted as TPs (white). Incorrectly predicted changed areas where no actual change exists are denoted as FPs (red). Actual changed areas missed by the model are denoted as FNs (blue). (a)–(c) Model predictions on WHU-CD dataset, GZ-CD dataset, and EGY-BCD dataset, respectively, highlighting generalization capability under domain shifts. "w/" represents the baseline model with our different loss function. And OURS means the baseline model with full of our method.

TABLE VI

ABLATION STUDIES. LEVIR-CD DATASET IS USED AS A TRAINING DATASET. THE DUAL-COVARIANCE STYLE-DEBIAS MECHANISM AND EVIDENCE-AWARE UNCERTAINTY ESTIMATION MODULE ARE EVALUATED. BEST RESULTS ARE SHOWN IN BOLD

L^1_{single}	L^2_{single}	L_{dual}	L_{Uncer}	Acc (%)	mIoU (%)	mF1 (%)
✗	✗	✗	✗	92.92	62.17	73.36
✓	✗	✗	✗	94.50	65.88	75.27
✓	✓	✗	✗	94.75	66.21	75.30
✓	✓	✓	✗	**94.92**	**67.83**	**77.35**
✗	✗	✗	✓	94.56	64.05	72.65
✓	✓	✓	✓	**95.11**	**68.63**	**78.08**

and improve generalization. Furthermore, the overall mIoU coefficient exceeds the baseline by 6.46% when using both dual-covariance style-debias mechanism compared and the evidence-aware uncertainty estimation module. It is indicated that the application of the GenerChange is useful and provides the best performance with an overall mIoU of 68.63% compared to the baseline.

Visualization in Fig. 3(a)–(c) verifies these findings. In (a), the WHU-CD dataset, baseline models confuse seasonal vegetation changes with structural changes, causing high uncertainty and FNs. Adding L_{single} educes these errors by distinguishing permanent structures. Integrating L_{Style} and L_{Uncer} further suppresses clustered FPs via style-debiasing and evidential learning. As shown in (b), fog causes aleatoric uncertainty and FPs. L_{Style} reduces FNs from irrigation-induced style shifts. The full model exhibits comprehensive performance to make both FP and FN to be less. This progressive error reduction visually demonstrates how epistemic uncertainty materializes as boundary-aligned FNs in unfamiliar environments, resolved through our content-preserving

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TABLE VII

PARAMETER SENSITIVITY ANALYSIS. LEVIR-CD DATASET IS USED AS A TRAINING DATASET. SENSITIVITY OF λ_1 AND λ_2 PARAMETERS (PARAM) EVALUATED ACROSS MULTIPLE DATASETS. BEST RESULTS ARE SHOWN IN BOLD, SECOND-BEST IN BLUE

Param		WHU-CD			GZ-CD			EGY-BCD			AVG		
λ_1	λ_2	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)
1.0	1.0	**97.63**	**77.22**	**85.64**	92.98	63.87	74.05	**94.71**	**64.81**	**74.56**	**95.11**	**68.63**	**78.08**
0.8	1.0	97.17	76.83	85.39	93.43	67.23	77.47	93.88	59.13	68.07	94.83	67.73	76.97
0.6	1.0	96.90	74.38	83.38	92.50	65.27	75.70	93.34	**63.63**	**73.68**	94.25	67.76	77.59
0.4	1.0	96.74	74.15	83.21	**93.62**	**68.60**	**78.78**	**94.18**	61.66	71.12	94.85	68.14	77.70
1.0	0.8	97.36	76.18	84.83	93.24	66.24	76.51	93.70	60.34	69.68	94.76	67.59	77.00
1.0	0.6	**97.45**	**77.90**	**86.22**	**93.64**	**68.95**	**79.11**	94.10	60.31	69.49	**95.06**	**69.06**	**78.27**
1.0	0.4	97.37	75.88	84.58	92.71	63.06	73.23	93.99	58.84	67.65	94.69	65.93	75.15

TABLE VIII

PARAMETER SENSITIVITY ANALYSIS. LEVIR-CD DATASET IS USED AS A TRAINING DATASET. SENSITIVITY OF ω_1 AND ω_2 PARAMETERS (PARAM) EVALUATED ACROSS MULTIPLE DATASETS

Param		WHU-CD			GZ-CD			EGY-BCD			AVG		
ω_1	ω_2	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)	Acc (%)	mIoU (%)	mF1 (%)
0.2	0.2	**97.63**	**77.22**	**85.64**	92.98	63.87	74.05	**94.71**	**64.81**	**74.56**	**95.11**	68.63	78.08
0.1	0.2	**97.50**	77.13	85.59	93.24	66.00	76.26	**94.37**	61.65	71.03	**95.04**	68.26	77.62
0.4	0.2	97.44	**77.31**	**85.74**	93.37	**68.24**	**78.49**	93.77	62.99	72.81	94.86	**69.51**	**79.01**
0.8	0.2	97.23	75.91	84.63	93.31	66.40	76.66	93.99	58.59	67.33	94.84	66.97	76.20
0.2	0.1	96.82	75.00	83.93	92.53	64.73	75.12	93.76	58.27	66.99	94.37	66.00	75.35
0.2	0.4	97.46	75.59	84.32	**93.57**	67.17	77.37	94.31	60.24	69.32	**95.11**	67.66	77.00
0.2	0.8	95.77	70.46	80.05	**94.24**	**72.95**	**82.65**	93.74	**63.46**	**73.36**	94.58	**68.96**	**78.69**

adaptation. For the EGY-BCD dataset in (c), desert texture and shape variations cause baseline FPs and FNs. The full model addresses both aleatoric and epistemic uncertainty, significantly reducing FPs.

3) Tradeoff of λ_1 and λ_2: To show the tradeoff of the parameters of $L_{\text{Style}} = \lambda_1 \cdot L_{\text{single}} + \lambda_2 \cdot L_{\text{dual}}$, we have supplemented an ablation study analyzing the weighting parameters λ_1 and λ_2, with results detailed in Table VII. Using the BIT-CD model with a ResNet-50 backbone, we train on the LEVIR-CD and test on EGY-BCD and GZ-CD. The experimental data reveal distinct tradeoff characteristics. When λ_1 decreases from 1.0 to 0.4, while fixing $\lambda_2 = 1.0$, average performance across three datasets declines by 0.26% Acc, 0.49% mIoU and 0.38% mF1. This confirms that suppressing style differences through L_{single} is essential to reduce aleatoric uncertainty, as insufficient weighting compromises domain-invariant feature learning. In contrast, reducing λ_2 from 1.0 to 0.4, with $\lambda_1 = 1.0$ initially, causes greater degradation, demonstrating the vital role L_{dual} in protecting semantic content during style-debias. In particular, at $\lambda_2 = 0.6$, performance surpasses the configuration $\lambda_1 = \lambda_2 = 1.0$ with an average mIoU increasing from 68.63% to 69.06%. This nonlinear relationship suggests that moderate relaxation of the dual-temporal covariance constraint prevents over-regularization of content features. That validates our theoretical claim that excessive style suppression damages structural information. The performance collapse at $\lambda_2 = 0.4$ further supports this, indicating that when content preservation is insufficiently weighted, the model fails to distinguish building changes from stylistic variations. This analysis quantitatively substantiates the adaptive balancing mechanism of our framework. L_{single} eliminates domain-specific noise while L_{dual} acts as a content guardian through cross-temporal feature covariance alignment, achieving optimal equilibrium at $\lambda_1 = 1.0$ and $\lambda_2 = 0.6$.

4) Tradeoff of ω_1 and ω_2: To show the tradeoff of the parameters of $L_{\text{Sing}} = \omega_1 \cdot L_{\text{Style}} + \omega_2 \cdot L_{\text{Uncer}}$, we conducted a comprehensive hyperparameter sensitivity analysis for the weighting coefficients ω_1 and ω_2, with detailed experimental results presented in Table VIII. This investigation reveals critical insights into the balance between style disentanglement and uncertainty optimization. The optimal configuration is achieved at $\omega_1 = 0.4$ and $\omega_2 = 0.2$, where our method achieves peak performance with 69.51% average mIoU and 79.01% mF1 across the datasets, representing improvements of 0.88% and 0.93%, respectively, over the baseline. Performance degradation occurs when ω_1 increases beyond 0.4 while maintaining $\omega_2 = 0.2$, evidenced by a 2.54% mIoU reduction at $\omega_1 = 0.8$ as the over-emphasis on style debiasing compromises uncertainty modeling. Similarly, reducing ω_2 below 0.2 severely impacts performance, with the $\omega_2 = 0.1$ configuration suffering 2.63% average mIoU degradation due to insufficient regularization of epistemic uncertainty. These findings quantitatively validate our theoretical framework. The dual-covariance mechanism requires moderate weighting, when $\omega_1 = 0.1$ to eliminate style variations without oversuppressing content features. And the evidence-aware module requires stronger regularization when $\omega_2 = 0.2$ to effectively correct probability distributions between domains. This optimal equilibrium enables robust cross-domain change detection where seasonal variations and class imbalance are simultaneously addressed through complementary loss components.

5) Effectiveness of Batch Size and Learning Rate: As illustrated in Fig. 4 and shown in Tables IX and X, we analyze the sensitivity of the proposed model to batch size and learning rate variations when trained on the LEVIR-CD dataset. The left picture reveals that the highest test mIoU is achieved with a batch size of 4, outperforming batch sizes 2 and 6, when the learning rate is fixed to 0.01. This is because smaller

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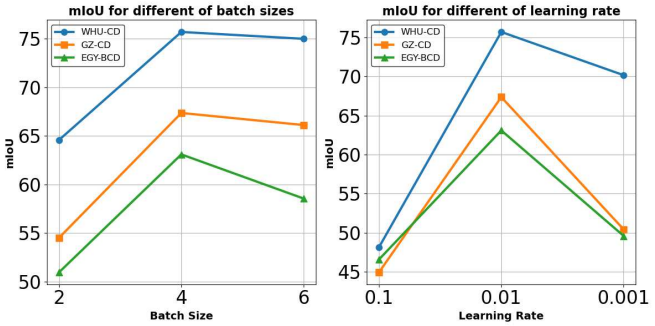


Fig. 4. Quantitative analysis of mIoU performance for the proposed model trained on the LEVIR-CD dataset (ResNet-50 backbone) and evaluated on three cross-domain test sets: WHU-CD, GZ-CD, and EGY-BCD. (Left) mIoU variation across batch sizes. (Right) mIoU trends under learning rates. Results demonstrate the cross-domain robustness of the model and highlight hyperparameter sensitivity in unseen scenarios, with optimal performance achieved at batch size 4 and learning rate 0.01.

TABLE IX

COMPARED TO DIFFERENT BATCH SIZE WITH QUANTITATIVE COMPARISON OF MIOU (%). BASED ON THE RESNET-50 BACKBONE, LEVIR IS USED AS A TRAINING DATASET. BEST RESULTS ARE SHOWN IN BOLD

Param Batch size	**WHU-CD** mIoU (%)	**GZ-CD** mIoU (%)	**EGY-BCD** mIoU (%)	**Avg** mIoU (%)
2	64.58	54.51	50.96	56.69
4	**75.69**	**67.36**	**63.09**	**68.72**
6	74.99	66.11	58.55	66.55

TABLE X

COMPARED TO DIFFERENT LEARNING RATE WITH QUANTITATIVE COMPARISON OF MIOU (%). BASED ON THE RESNET-50 BACKBONE, LEVIR IS USED AS A TRAINING DATASET. BEST RESULTS ARE SHOWN IN BOLD

Param Learning rate	**WHU-CD** mIoU (%)	**GZ-CD** mIoU (%)	**EGY-BCD** mIoU (%)	**Avg** mIoU (%)
0.1	48.14	44.94	46.58	46.56
0.01	**75.69**	**67.36**	**63.09**	**68.72**
0.001	70.16	50.44	49.59	56.73

batches enhance gradient diversity through noisier updates, thereby improving generalization to unseen domains. However, excessively small batches degrade performance due to unstable gradient estimates and incomplete training dynamics. The right picture demonstrates that a learning rate of 0.01 yields optimal performance, surpassing both higher (0.1) and lower (0.001), when the batch size is fixed at 4. This phenomenon reflects that excessively high rates induce oscillatory behavior, while overly low rates hinder escape from shallow local minima. Our results suggest that the moderate batch size and learning rate balance these competing objectives for robustness.

V. CONCLUSION

In this article, we introduce a novel GenerChange to detect remote sensing image change from different domains. The method proposes the dual-covariance style-debias mechanism to remove the style information of dual-temporal images, which can make the model adaptively learn the content information. Furthermore, the evidence-aware uncertainty estimation module is designed to collect evidence to reduce epistemic uncertainty and make the model learn the domain-invariant features to improve the generalization ability of the model. Extensive experimental results show the effectiveness and high performance in DG RSCD tasks. However, our current framework remains dependent on substantial source-domain annotated data to achieve better performance. This limitation may constrain real-world applicability given the frequent scarcity of comprehensively labeled datasets. Thus, DG change detection under few-shot learning paradigms remains an unresolved challenge. Furthermore, while our cross-domain methodology effectively addresses challenges in optical images, its generalization ability across heterogeneous sensor modalities remains underexplored. Therefore, how to improve the generalization ability of the model to the heterogeneous sensor modalities is significant. In the future, we will develop a robust DG change detection model with the lack of labeled data. Second, we will architect the heterogeneous DG method to enhance model transferability across diverse sensing technologies, including synthetic aperture radar, multispectral, and hyperspectral images.

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